

Term Life
Calculation Specification

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INTRODUCTION

This document provides a comprehensive overview of the formulas for Inforce Term products. The business rules are not considered in this document. It is divided into 4 parts:

- 1) **Duration:** This section details the formulas of the various durations used in the calculations
- 2) **Input Setup:** This section details the inputs we get from Capsil, and the various calculations related to those inputs.
- 3) **Report:** This section details the formulas for each variable appearing in the Illustration Report (projection values). For report purposes, the projections are calculated as of the date of the next policy anniversary.

The coverages included in the calculations are all the term products and their additional guarantees. The following table summarizes which type of coverage must be considered for each type of calculation. (Green = Consider the coverage in the calculation; Grey = Don’t consider the coverage in the calculation)

#	Calculation	Term Products	Additional Coverages
1	Duration		
2	Age		
3	Total Premium		
4	Life Insurance Premium		
5	Cumulative Premiums		
6	Total Extra Premium		
7	Life Insurance Amount		
8	Additional Coverages Premium		
9	Coverage Net Premium Reserve		
10	Net Amount at Risk (NAAR)		
11	Net Cost of Pure Insurance (NCPI)		
12	Cumulative NCPI		
13	ACB Coverage Cost		
14	Cumulative ACB Coverage Cost		
15	Adjusted Cost Basis (ACB)		

1 Duration

$iDur$ = Duration, $iCov$ = Coverage, $iInsured$ = Insured

1.1 End of Coverage Projection

$$End\ of\ Coverage\ Projection_{(iCov)} = End\ of\ Coverage\ Projection\ Without\ Adjustment_{(iCov)} + Coverage\ Projection\ Adjustment_{(iCov)}$$

1.1.1 End of Coverage Projection Without Adjustment

$$End\ of\ Coverage\ Projection\ Without\ Adjustment_{(iCov)} = YEAR\left(Coverage\ Expiration\ Date_{(iCov)}\right) - YEAR\left(Coverage\ Issue\ Date_{(iCov)}\right)$$

1.1.1.1 Coverage Expiration Date Refer to definition

1.1.1.2 Coverage Issue Date Refer to definition

1.1.2 Coverage Projection Adjustment

$$Coverage\ Projection\ Adjustment_{(iCov)} = \begin{cases} 0 & \text{if Policy vs Coverage Offset}_{(iCov)} = 0 \\ 1 & \text{otherwise} \end{cases}$$

1.1.2.1 Policy vs Coverage Offset Refer to definition

1.2 End of Illustration Projection

$$End\ of\ Illustration\ Projection = \max_{1 \leq iCov \leq Nb\ of\ Cov.\ on\ illustration} \left\{ End\ of\ Coverage\ Projection_{(iCov)} \right\}$$

1.2.1 End of Coverage Projection

Refer to previous definition

1.3 End of Premium Payment

$$End\ of\ Premium\ Pmt_{(iCov)} = End\ of\ Premium\ Pmt\ Without\ Adjustment_{(iCov)} + Coverage\ Projection\ Adjustment_{(iCov)}$$

1.3.1 End of Premium Pmt Without Adjustment

If: *End of Premium Type*_(iCov) = *A or B*, then

$$End\ of\ Premium\ Pmt\ Without\ Adjustment_{(iCov)} = YEAR\left(Coverage\ Paid\ up\ Date_{(iCov)}\right) - YEAR\left(Coverage\ Issue\ Date_{(iCov)}\right)$$

1.3.1.1 Coverage Paid-up Date Refer to definition

1.3.1.2 Coverage Issue Date Refer to definition

If: *End of Premium Type*_(iCov) = *C*, then

$$End\ of\ Premium\ Pmt\ Without\ Adjustment_{(iCov)} = \begin{cases} End\ of\ Premium\ Value_{(iCov)} & \text{if } Expiration\ Type_{(iCov)} = A \\ End\ of\ Premium\ Value_{(iCov)} - Maximum\ Age_{(iCov)} & \text{otherwise} \end{cases}$$

1.3.1.3 Expiration Type

$$Expiration\ Type_{(iCov)} = \begin{cases} A & : \text{Coverage never ends} \\ B & : \text{Coverage ends at a predetermined number of years} \\ C & : \text{Coverage ends at a predetermined age} \end{cases}$$

See Product Characteristics document to associate to the right product.

1.3.1.4 End of Premium Value

$$End\ of\ Premium\ Value_{(iCov)} = \begin{cases} MIN\left\{100 ; Expiration\ Value_{(iCov)}\right\} & \text{if } End\ of\ Premium\ Type_{(iCov)} = C \\ End\ of\ Premium\ (Age\ or\ Duration)_{(iCov)} & \text{otherwise} \end{cases}$$

1.3.1.4.1 End of Premium (Age or Duration)

$$End\ of\ Premium\ (Age\ or\ Duration)_{(iCov)} = \text{Age or number of years at which the premium ends}$$

1.3.1.4.2 Expiration Value

$$Expiration\ Value_{(iCov)} = \begin{cases} 100 & \text{if } Expiration\ Type_{(iCov)} = A \\ End\ of\ Coverage\ (Age\ or\ Duration)_{(iCov)} & \text{otherwise} \end{cases}$$

1.3.1.4.3 End of Coverage (Age or Duration)

$$End\ of\ Coverage\ (Age\ or\ Duration)_{(iCov)} = \text{Age or number of years at which the coverage ends}$$

See Product Characteristics document to associate to the right product.

1.3.1.4.4 End of Premium Type

$$End\ of\ Premium\ Type_{(iCov)} = \begin{cases} A & : \text{Premiums end in a predetermined number of years} \\ B & : \text{Premiums end at a predetermined age} \\ C & : \text{Premium never ends} \end{cases}$$

See Product Characteristics document to associate to the right product.

1.3.1.5 Expiration Value Refer to previous definition

1.3.1.6 Maximum Age

$$Maximum\ Age_{(iCov)} = \begin{cases} \text{Age of the insured (from Capsil)} & \text{for Individual Coverage} \\ \text{Equivalent age (from Capsil)} & \text{for Joint Coverage of the equivalent age type (ex: Permanent life products)} \\ \text{Age of the oldest insured on the coverage (from Capsil)} & \text{for Joint Coverage with no equivalent age (ex: Term life products)} \end{cases}$$

1.3.2 Coverage Projection Adjustment

Refer to previous definition

1.4 End of Fiscal Projection

$$End\ of\ Fiscal\ Projection_{(iCov,iInsured)} = \begin{cases} \Omega_{(iCov)} - Age_{(iCov,iInsured)} & \text{if } Expiration\ Type_{(iCov)} = A \text{ or } C \\ \Omega_{(iCov)} & \text{otherwise} \end{cases}$$

→ Additional coverages are not included in the coverages calculated

1.4.1 Expiration Type

Refer to previous definition

1.4.2 Omega

$$\Omega_{(iCov)} = \begin{cases} 106 & \text{for Permanent products} \\ Expiration\ Value_{(iCov)} & \text{for Term products} \end{cases}$$

1.4.2.1 Expiration Value Refer to previous definition

1.4.3 Age

$$Age_{(iCov,iInsured)} = \text{Age for each of the insured on the coverage (from Capsil)}$$

2 INPUT SETUP: For the Policy

2.1 Data from CAPSIL

2.1.1 Policy Issue Date

$$Policy\ Issue\ Date = \text{Issue date of the policy (from CAPSIL)}$$

$$\rightarrow YEAR(Policy\ Issue\ Date) = \text{Year of the Policy Issue Date}$$

$$\rightarrow MONTH(Policy\ Issue\ Date) = \text{Month of the Policy Issue Date}$$

$$\rightarrow DAY(Policy\ Issue\ Date) = \text{Day of the Policy Issue Date}$$

2.1.2 Value As of Date

$$Value\ As\ of\ Date = \text{Date on which the data is retrieved from capsil (from CAPSIL)}$$

$$\rightarrow YEAR(Value\ As\ of\ Date) = \text{Year of the Value As of Date}$$

$$\rightarrow MONTH(Value\ As\ of\ Date) = \text{Month of the Value As of Date}$$

$$\rightarrow DAY(Value\ As\ of\ Date) = \text{Day of the Value As of Date}$$

2.1.3 Premiums Paid to Date

$$Premiums\ Paid\ to\ Date = \text{Premiums have been paid up to this date (from CAPSIL)}$$

$$\rightarrow YEAR(Premiums\ Paid\ to\ Date) = \text{Year of the Premiums Paid to Date}$$

$$\rightarrow MONTH(Premiums\ Paid\ to\ Date) = \text{Month of the Premiums Paid to Date}$$

$$\rightarrow DAY(Premiums\ Paid\ to\ Date) = \text{Day of the Premiums Paid to Date}$$

2.1.4 Total Premiums Paid

$$Total\ Premiums\ Paid = \text{Total cumulative premiums already paid for the policy (from CAPSIL)}$$

2.1.5 Policy Cumulative NCPI

$$Policy\ Cumulative\ NCPI = \text{Total cumulative Net Cost of pure Insurance to date for the policy (from CAPSIL, the value is as of last December 31)}$$

2.1.6 Policy ACB

$$Policy\ ACB = \text{Adjusted Cost Basis to date for the policy (from CAPSIL)}$$

2.2 Next Policy Anniversary

$$Next\ Policy\ Anniversary = DATE\{YEAR(Next\ Policy\ Anniversary) ; MONTH(Policy\ Issue\ Date) ; DAY(Policy\ Issue\ Date)\}$$

2.2.1 Policy Issue Date

Refer to previous definition

2.2.2 YEAR (Next Policy Anniversary)

$$YEAR(Next Policy Anniversary) = \begin{cases} YEAR(Value As of Date) + 1 & \text{if } MONTH(Policy Issue Date) < MONTH(Value As of Date) \\ YEAR(Value As of Date) + 1 & \text{if } MONTH(Policy Issue Date) = MONTH(Value As of Date) \text{ and } DAY(Policy Issue Date) \leq DAY(Value As of Date) \\ YEAR(Value As of Date) & \text{otherwise} \end{cases}$$

2.2.2.1 Value As of Date *Refer to previous definition*

2.3 Nbr Premiums Left Until Next Policy Anniversary

If payment mode is Annual, then

$$Nbr Premiums Until Next Policy Anniv = MIN \left\{ 1 ; \frac{Nbr of days between (Next Policy Anniversary AND Premiums Paid to Date)}{365} \right\}$$

*For the number of days between 2 dates, we assume a 365-day year, with February considered to have 28 days.

If payment mode is Monthly, then

$$Nbr Premiums Until Next Policy Anniv = \begin{cases} MONTH(Next Policy Anniversary) - MONTH(Premiums Paid to Date) & \text{if } MONTH(Next Policy Anniversary) \geq MONTH(Premiums Paid to Date) \\ MONTH(Next Policy Anniversary) - MONTH(Premiums Paid to Date) + 12 & \text{if } MONTH(Next Policy Anniversary) < MONTH(Premiums Paid to Date) \end{cases}$$

2.3.1 Next Policy Anniversary

Refer to previous definition

2.3.2 Premiums Paid to Date

Refer to previous definition

2.4 Policy 1st Duration

$$Policy 1st Duration = YEAR(Value As of Date) - YEAR(Policy Issue Date) + Policy Duration Adjustment$$

2.4.1 Value As of Date

Refer to previous definition

2.4.2 Policy Issue Date

Refer to previous definition

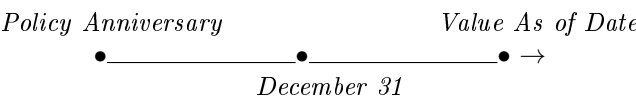
2.4.3 Policy Duration Adjustment

$$Policy Duration Adjustment = \begin{cases} 0 & \text{if } MONTH(Value As of Date) < MONTH(Policy Issue Date) \\ 0 & \text{if } MONTH(Value As of Date) = MONTH(Policy Issue Date) \text{ AND } DAY(Value As of Date) < DAY(Policy Issue Date) \\ 1 & \text{otherwise} \end{cases}$$

2.5 Policy Dec 31st Indicator

The indicator is calculated to help determine whether the NCPI has already been deducted from the ACB or not.

If the *Value As of Date* is after December 31 and the *Policy Annviersary* is before December 31, then the NCPI has already been deducted:



$$Policy Dec 31st Indicator = \begin{cases} 1 & \text{if } MONTH(Value As of Date) < MONTH(Policy Issue Date) \\ 1 & \text{if } MONTH(Value As of Date) = MONTH(Policy Issue Date) \text{ AND } DAY(Value As of Date) < DAY(Policy Issue Date) \\ 0 & \text{otherwise} \end{cases}$$

2.5.1 Policy Issue Date

Refer to previous definition

2.5.2 Value As of Date

Refer to previous definition

3 INPUT SETUP: For the coverages

3.1 Data from CAPSIL

3.1.1 Coverage Issue Date

$Coverage\ Issue\ Date_{(iCov)} =$ Issue date of the coverage (*from CAPSIL for each coverages on the policy*)

$$\rightarrow YEAR\left(Coverage\ Issue\ Date_{(iCov)}\right) = Year\ of\ the\ Coverage\ Issue\ Date$$

$$\rightarrow MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) = Month\ of\ the\ Coverage\ Issue\ Date$$

$$\rightarrow DAY\left(Coverage\ Issue\ Date_{(iCov)}\right) = Day\ of\ the\ Coverage\ Issue\ Date$$

3.1.2 Coverage Expiration Date

$Coverage\ Expiration\ Date_{(iCov)} =$ Expiration date of the coverage (*from CAPSIL for each coverages on the policy*)

$$\rightarrow YEAR\left(Coverage\ Expiration\ Date_{(iCov)}\right) = Year\ of\ the\ Coverage\ Expiration\ Date$$

$$\rightarrow MONTH\left(Coverage\ Expiration\ Date_{(iCov)}\right) = Month\ of\ the\ Coverage\ Expiration\ Date$$

$$\rightarrow DAY\left(Coverage\ Expiration\ Date_{(iCov)}\right) = Day\ of\ the\ Coverage\ Expiration\ Date$$

3.1.3 Coverage Paid-up Date

$Coverage\ Paid\ up\ Date_{(iCov)} =$ End of premium date of the coverage (*from CAPSIL for each coverages on the policy*)

$$\rightarrow YEAR\left(Coverage\ Paid\ up\ Date_{(iCov)}\right) = Year\ of\ the\ Coverage\ Paid\ up\ Date$$

$$\rightarrow MONTH\left(Coverage\ Paid\ up\ Date_{(iCov)}\right) = Month\ of\ the\ Coverage\ Paid\ up\ Date$$

$$\rightarrow DAY\left(Coverage\ Paid\ up\ Date_{(iCov)}\right) = Day\ of\ the\ Coverage\ Paid\ up\ Date$$

3.2 Coverage 1st Duration

$$Coverage\ 1st\ Duration_{(iCov)} = YEAR(Value\ As\ of\ Date) - YEAR\left(Coverage\ Issue\ Date_{(iCov)}\right) + Coverage\ Duration\ Adjustment_{(iCov)}$$

3.2.1 Value As of Date

Refer to previous definition

3.2.2 Coverage Issue Date

Refer to previous definition

3.2.3 Coverage Duration Adjustment

$$Coverage\ Duration\ Adjustment_{(iCov)} = \begin{cases} 0 & \text{if } MONTH(Value\ As\ of\ Date) < MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \\ 0 & \text{if } MONTH(Value\ As\ of\ Date) = MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \text{ and } DAY(Value\ As\ of\ Date) < DAY\left(Coverage\ Issue\ Date_{(iCov)}\right) \\ 1 & \text{otherwise} \end{cases}$$

3.3 Premium Rate Duration

$$Premium\ Rate\ Duration_{(iCov)} = YEAR(Premiums\ Paid\ to\ Date) - YEAR\left(Coverage\ Issue\ Date_{(iCov)}\right) + Premium\ Duration\ Adjustment_{(iCov)}$$

3.3.1 Premiums Paid to Date

Refer to previous definition

3.3.2 Coverage Issue Date

Refer to previous definition

3.3.3 Premium Duration Adjustment

$$Premium\ Duration\ Adjustment_{(iCov)} = \begin{cases} 0 & \text{if } MONTH(Premiums\ Paid\ to\ Date) < MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \\ 0 & \text{if } MONTH(Premiums\ Paid\ to\ Date) = MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \text{ and } DAY(Premiums\ Paid\ to\ Date) < DAY\left(Coverage\ Issue\ Date_{(iCov)}\right) \\ 1 & \text{otherwise} \end{cases}$$

3.4 Policy vs Coverage Anniversary Offset

If payment mode is Annual, then

$$Policy\ vs\ Coverage\ Offset_{(iCov)} = Nbrs\ of\ days\ between\ \left(Adjusted\ Coverage\ Issue\ Date_{(iCov)}\ AND\ Policy\ Issue\ Date\right)$$

*For the number of days between 2 dates, we assume a 365-day year, with February considered to have 28 days.

If payment mode is Monthly, then

$$Policy\ vs\ Coverage\ Offset_{(iCov)} = \begin{cases} MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) - MONTH(Policy\ Issue\ Date) & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \geq MONTH(Policy\ Issue\ Date) \\ MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) - MONTH(Policy\ Issue\ Date) + 12 & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) < MONTH(Policy\ Issue\ Date) \end{cases}$$

3.4.1 Adjusted Coverage Issue Date

$$Adjusted\ Coverage\ Issue\ Date_{(iCov)} = DATE\left\{Adjusted\ Coverage\ Issue\ Year_{(iCov)} ; MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) ; DAY\left(Coverage\ Issue\ Date_{(iCov)}\right)\right\}$$

3.4.1.1 Adjusted Coverage Issue Year

$$Adjusted\ Coverage\ Issue\ Year_{(iCov)} = \begin{cases} YEAR(Policy\ Issue\ Date) & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) \geq MONTH(Policy\ Issue\ Date) \\ YEAR(Policy\ Issue\ Date) + 1 & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) < MONTH(Policy\ Issue\ Date) \end{cases}$$

3.4.2 Policy Issue Date

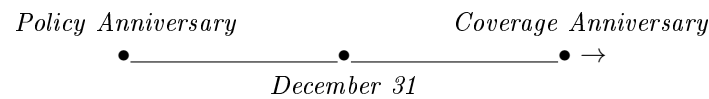
Refer to previous definition

3.4.3 Coverage Issue Date

Refer to previous definition

3.5 Coverage Dec 31st Indicator

The indicator is calculated to help determine whether the calculation of the NCPI should be deferred because the *Policy Anniversary* is before December 31 and the *Coverage Anniversary* is after December 31.



$$Coverage\ Dec\ 31st\ Indicator_{(iCov)} = \begin{cases} -1 & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) < MONTH(Policy\ Issue\ Date) \\ -1 & \text{if } MONTH\left(Coverage\ Issue\ Date_{(iCov)}\right) = MONTH(Policy\ Issue\ Date) \text{ and } DAY\left(Coverage\ Issue\ Date_{(iCov)}\right) < DAY(Policy\ Issue\ Date) \\ 0 & \text{otherwise} \end{cases}$$

3.5.1 Policy Issue Date

Refer to previous definition

3.5.2 Coverage Issue Date

Refer to previous definition

4 REPORT: Age

For the global illustration / policy:

$$Age_{(iDur)} = Illustration \ Minimum \ Age + iDur$$

→ Where iDur is between Policy 1st Duration and End of Illustration Projection
For the coverages:

$$Age_{(iCov, iDur)} = Coverage \ Minimum \ Age_{(iCov)} + iDur$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

4.1 Illustration Minimum Age

$$Illustration \ Minimum \ Age = \min_{1 \leq iCov \leq Nb \ of \ Cov. \ on \ illustration} \left\{ Coverage \ Minimum \ Age_{(iCov)} \right\}$$

4.2 Coverage Minimum Age

$$Coverage \ Minimum \ Age_{(iCov)} = Coverage \ Issue \ Age_{(iCov)} + Coverage \ 1st \ Duration_{(iCov)}$$

4.2.1 Coverage Issue Age

$$Coverage \ Issue \ Age_{(iCov)} = Issue \ age \ of \ the \ coverage \ (from \ CAPSIL \ for \ each \ coverages \ on \ the \ policy)$$

4.2.2 Coverage 1st Duration

Refer to previous definition

5 REPORT: Total Premium

$$Total \ Premium_{(Policy \ 1st \ Duration + iDur - 1)} = \sum_{iCov=1}^{NbCov} Total \ Interpolated \ Premium_{(iCov, \ Cov \ 1st \ Duration_{(iCov)} + iDur - 1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)
→ Coverage 1st Duration

5.1 Total Interpolated Premium

→ If Payment Frequency is **Annual**

⇒ If $iDur = Coverage \ 1st \ Duration_{(iCov)}$, then

$$\Rightarrow \textbf{Total Interpolated Premium}_{(iCov, iDur)} = Nbr \ Premiums \ Until \ Next \ Policy \ Anniv * \left[\left(\frac{Policy \ vs \ Coverage \ Offset_{(iCov)}}{365} \right) * Modal \ Premium_{(iCov, iDur-1)} + \left(1 - \frac{Policy \ vs \ Coverage \ Offset_{(iCov)}}{365} \right) * Modal \ Premium_{(iCov, iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

⇒ If $iDur > Coverage \ 1st \ Duration_{(iCov)}$, then

$$\Rightarrow \textbf{Total Interpolated Premium}_{(iCov, iDur)} = \left[\left(\frac{Policy \ vs \ Coverage \ Offset_{(iCov)}}{365} \right) * Modal \ Premium_{(iCov, iDur-1)} \right] + \left[\left(1 - \frac{Policy \ vs \ Coverage \ Offset_{(iCov)}}{365} \right) * Modal \ Premium_{(iCov, iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

→ If Payment Frequency is **Monthly**

⇒ If $iDur = Coverage \ 1st \ Duration_{(iCov)}$, then

a) If $Nbr \ Premiums \ Until \ Next \ Policy \ Anniv \leq 12 - Policy \ vs \ Coverage \ Offset_{(iCov)}$, then

$$\Rightarrow \textbf{Total Interpolated Premium}_{(iCov, iDur)} = Nbr \ Premiums \ Until \ Next \ Policy \ Anniv * Modal \ Premium_{(iCov, iDur)}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

b) Otherwise

$$\Rightarrow \textbf{Total Interpolated Premium}_{(iCov, iDur)} = \left[\left(12 - Policy \ vs \ Coverage \ Offset_{(iCov)} \right) * Modal \ Premium_{(iCov, iDur)} \right] + \left[\left(Nbr \ Premiums \ Until \ Next \ Policy \ Anniv - \left(12 - Policy \ vs \ Coverage \ Offset_{(iCov)} \right) \right) * Modal \ Premium_{(iCov, iDur-1)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

⇒ If $iDur > Coverage \ 1st \ Duration_{(iCov)}$, then

$$\Rightarrow \textbf{Total Interpolated Premium}_{(iCov, iDur)} = \left[Policy \ vs \ Coverage \ Offset_{(iCov)} * Modal \ Premium_{(iCov, iDur-1)} \right] + \left[\left(12 - Policy \ vs \ Coverage \ Offset_{(iCov)} \right) * Modal \ Premium_{(iCov, iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.1.1 Nbr Premiums Left Until Next Policy Anniversary

Refer to previous definition

5.1.2 Policy vs Coverage Anniversary Offset

Refer to previous definition

5.2 Modal Premium

$$Modal\ Premium_{(iCov,iDur)} = ROUND \left\{ \left(Cost\ of\ insurance_{(iCov,iDur)} + Percentage\ Extra\ Premium_{(iCov,iDur)} + Dollar\ Extra\ Premium_{(iCov,iDur)} \right) * Modal\ factor_{(iCov,iDur)} ; 2 \right\} + ROUND \left\{ Coverage\ fee_{(iCov,iDur)} * Modal\ factor_{(iCov,iDur)} ; 2 \right\}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.2.1 Cost of Insurance

$$Cost\ of\ Insurance_{(iCov, iDur)} = ROUND \left\{ ROUND \left\{ TRUNC \left\{ Coverage\ premium\ rate_{(iCov, iDur)} ; 6 \right\} * ROUND \left\{ \frac{Insurance\ Amount_{(iCov, iDur)}}{Unit\ value} ; 5 \right\} ; 2 \right\} * PREM_ADJ\%_{(iCov,iDur)} + PREM_ADJ\$_{(iCov,iDur)} ; 2 \right\}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.2.1.1 Coverage Premium Rate

$$Coverage\ premium\ rate_{(iCov,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Premium\ rate_{(iCov,iInsured,iDur)}$$

$$Premium\ rate_{(iCov,iInsured,iDur)} = \text{Premium rate for each insured (rate per 1000)}$$

→ Where iDur is between Premium Rate Duration and associated End of Premium rate table

For term products, there are 2 set of rates: Individual and JFTD

- **If the coverage type is Individual: Use Individual rates**
- **If the coverage type is Joint: Use JFTD rates. The rates for each insured person must be added together to obtain the overall rate for the coverage**
- **If there is a discount (if MCD discount applies: a flag will be passed from Capsil): Use the next higher band of the Individual rates**

5.2.1.2 Insurance Amount

$$Insurance\ Amount_{(iCov,iDur)} = \begin{cases} \text{Sum of all annual premiums for all coverages except the DW coverages} & \text{if DW coverage} \\ \text{Coverage Insurance Amount (from Capsil at first but can be modified)} & \text{otherwise} \end{cases}$$

→ Where DW = Disability Waiver premium

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.2.1.3 Unit value

$$Unit\ value = \begin{cases} 100 & \text{if coverage is DW or Long Term Care Advance} \\ 1000 & \text{otherwise} \end{cases}$$

→ Where DW = Disability Waiver premium

5.2.1.4 PREM_ADJ%

$$PREM_ADJ\%_{(iCov,iDur)} = \% \text{ of Premium Adjustment (from CAPSIL)}$$

→ Where iDur is between Premium Rate Duration and MIN (End of Premium Payment ; PREM_ADJ% Remaining Duration)

5.2.1.4.1 PREM_ADJ% Remaining Duration

$$PREM_ADJ\% \text{ Remaining Duration}_{(iCov)} = PREM_ADJ\% \text{ Duration}_{(iCov)} - Premium\ Rate\ Duration_{(iCov)} + 1$$

5.2.1.4.2 PREM_ADJ% Duration

$$PREM_{ADJ\% \text{ Duration}_{(iCov)}} = \text{Duration of the Premium Adjustment (from Capsil)}$$

5.2.1.4.3 Premium Rate Duration Refer to previous definition

5.2.1.5 PREM_ADJ\$

$$PREM_ADJ\$_{(iCov,iDur)} = \text{Premium Adjustment in \$ (field available in Special Quotes, will not be passed from Capsil, presume 0)}$$

→ Where iDur is between Premium Rate Duration and End of Coverage Projection

5.2.2 Percentage Extra Premium

$$Percentage\ Extra\ Premium_{(iCov,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Insured\ Percentage\ Extra\ Premium_{(iCov,iInsured,iDur)}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

$$Insured\ Percentage\ Extra\ Premium_{(iCov, iDur)} = ROUND \left\{ ROUND \left\{ TRUNC \left\{ Extra\ premium\ rate_{(iCov, iInsured,iDur)} ; 6 \right\} * Extra\ premium\ \%_{(iCov,iInsured)} * ROUND \left\{ \frac{Insurance\ Amount_{(iCov,iDur)}}{Unit\ value} ; 5 \right\} ; 2 \right\} * PREM_ADJ\%_{(iCov,iDur)} ; 2 \right\}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.2.2.1 Extra Premium Rate

$$Extra\ premium\ rate_{(iCov,iInsured,iDur)} = \begin{cases} \text{Substandard Premium rate for each insured} & \text{if Permanent product} \\ \text{Standard Premium rate for each insured} & \text{if Term product} \end{cases}$$

→ Where iDur is between Premium Rate Duration and associated End of Premium rate table

For term products, there are 2 set of rates: Individual and JFTD

- **If the coverage type is Individual: Use Individual rates**
- **If the coverage type is Joint: Use JFTD rates. The rates for each insured person must be added together to obtain the overall rate for the coverage**
- **If there is a discount (if MCD discount applies): Use the next higher band of the Individual rates**

5.2.2.2 Extra Premium %

$$Extra\ premium\ \%_{(iCov,iInsured)} = \% \text{ of the permanent extra premium input for each insured (from Capsil)}$$

→ For joint protections with equivalent age type (permanent products), only one extra premium is calculated.

5.2.2.3 Insurance Amount Refer to previous definition

5.2.2.4 Unit value Refer to previous definition

5.2.2.5 PREM_ADJ% Refer to previous definition

5.2.3 Dollar Extra Premium

$$Dollar\ Extra\ Premium_{(iCov,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Insured\ Dollar\ Extra\ Premium_{(iCov,iInsured,iDur)}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

$$Insured\ Dollar\ Extra\ Premium_{(iCov,iInsured,iDur)} = ROUND \left\{ Dollar\ Extra\ Premium\ Rate_{(iCov,iInsured,iDur)} * ROUND \left\{ \frac{Insurance\ Amount_{(iCov,iDur)}}{Unit\ value} ; 5 \right\} ; 2 \right\}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

5.2.3.1 Dollar Extra Premium Rate

$$Dollar\ Extra\ Premium\ Rate_{(iCov,iInsured,iDur)} = \text{Dollar extra premium rate for each insured (rate per 1000) (from Capsil)}$$

→ Where iDur is between Premium Rate Duration and MIN (End of Premium Payment ; Dollar Extra Premium Remaining Duration)

5.2.3.1.1 Dollar Extra Premium Remaining Duration

$$Dollar\ Extra\ Premium\ Remaining\ Duration_{(iCov,iInsured)} = Dollar\ Extra\ Premium\ Duration_{(iCov,iInsured)} - Premium\ Rate\ Duration_{(iCov)} + 1$$

5.2.3.1.2 Dollar Extra Premium Duration

$$Dollar\ Extra\ Premium\ Duration_{(iCov,iInsured)} = \text{Duration of the extra premium in \$ (from Capsil)}$$

5.2.3.1.3 Premium Rate Duration Refer to previous definition

5.2.3.2 Insurance Amount Refer to previous definition

5.2.3.3 Unit value Refer to previous definition

5.2.4 Modal Factor

$Modal\ factor_{(iCov)} = \text{Modal factor of the coverage}$ (see Product characteristics document)

5.2.5 Coverage Fee

$Coverage\ fee_{(iCov, iDur)} = \text{Fee for each coverage}$ (from Capsil)

→ Where iDur is between Premium Rate Duration and End of Premium Payment

6 REPORT: Life Insurance Premium

$$Life\ Insurance\ Premium_{(Policy\ 1^{st}\ Duration + iDur - 1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov, Cov\ 1^{st}\ Duration_{(iCov)} + iDur - 1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Policy 1st Duration and Coverage 1st Duration

→ Only life insurance coverages are included in this calculation

6.1 Total Interpolated Premium

Refer to previous definition

Considering only life insurance coverages.

7 REPORT: Cumulative Premiums

$$Cumulative\ Premiums_{(iDur)} = \begin{cases} Total\ Premiums\ Paid + Total\ Premium_{(iDur)} & \text{if } iDur = Policy\ 1^{st}\ Duration \\ Cumulative\ Premiums_{(iDur-1)} + Total\ Premium_{(iDur)} & \text{if } iDur > Policy\ 1^{st}\ Duration \end{cases}$$

→ Where iDur is between Policy 1st Duration and End of Illustration Projection

7.1 Total Premiums Paid

Refer to previous definition

7.2 Policy 1st Duration

Refer to previous definition

7.3 Total Premium

Refer to previous definition

8 REPORT: Total Extra Premium

$$Total\ Extra\ Premium_{(Policy\ 1^{st}\ Duration + iDur - 1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Extra\ Premium_{(iCov, Cov\ 1^{st}\ Duration_{(iCov)} + iDur - 1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

8.1 Total Interpolated Extra Premium

⇒ If Payment Frequency is **Annual**

⇒ If $iDur = Coverage\ 1^{st}\ Duration_{(iCov)}$, then

$$Total\ Interpolated\ Extra\ Premium_{(iCov, iDur)} = Nbr\ Premiums\ Until\ Next\ Policy\ Anniv * \left[\left(\frac{Policy\ vs\ Coverage\ Offset_{(iCov)}}{365} \right) * Modal\ Extra\ Premium_{(iCov, iDur-1)} + \left(1 - \frac{Policy\ vs\ Coverage\ Offset_{(iCov)}}{365} \right) * Modal\ Extra\ Premium_{(iCov, iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

⇒ If $iDur > Coverage\ 1^{st}\ Duration_{(iCov)}$, then

$$Total\ Interpolated\ Extra\ Premium_{(iCov, iDur)} = \left[\left(\frac{Policy\ vs\ Coverage\ Offset_{(iCov)}}{365} \right) * Modal\ Extra\ Premium_{(iCov, iDur-1)} \right] + \left[\left(1 - \frac{Policy\ vs\ Coverage\ Offset_{(iCov)}}{365} \right) * Modal\ Extra\ Premium_{(iCov, iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection
⇒ If Payment Frequency is **Monthly**
⇒ If $iDur = Coverage\ 1^{st}\ Duration_{(iCov)}$, then
a) If $Nbr\ Premiums\ Until\ Next\ Policy\ Anniv \leq 12 - Policy\ vs\ Coverage\ Offset_{(iCov)}$, then

$$Total\ Interpolated\ Extra\ Premium_{(iCov,\ iDur)} = Nbr\ Premiums\ Until\ Next\ Policy\ Anniv * Modal\ Extra\ Premium_{(iCov,iDur)}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection
b) Otherwise

$$Total\ Interpolated\ Extra\ Premium_{(iCov,\ iDur)} = \left[\left(12 - Policy\ vs\ Coverage\ Offset_{(iCov)} \right) * Modal\ Extra\ Premium_{(iCov,iDur)} \right] + \left[\left(Nbr\ Premiums\ Until\ Next\ Policy\ Anniv - \left(12 - Policy\ vs\ Coverage\ Offset_{(iCov)} \right) \right) * Modal\ Extra\ Premium_{(iCov,iDur-1)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection
⇒ If $iDur > Coverage\ 1^{st}\ Duration_{(iCov)}$, then

$$Total\ Interpolated\ Extra\ Premium_{(iCov,\ iDur)} = \left[Policy\ vs\ Coverage\ Offset_{(iCov)} * Modal\ Extra\ Premium_{(iCov,iDur-1)} \right] + \left[\left(12 - Policy\ vs\ Coverage\ Offset_{(iCov)} \right) * Modal\ Extra\ Premium_{(iCov,iDur)} \right]$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

8.1.1 Nbr Premiums Left Until Next Policy Anniversary

Refer to previous definition

8.1.2 Policy vs Coverage Anniversary Offset

Refer to previous definition

8.2 Modal Extra Premium

$$Modal\ Extra\ Premium_{(iCov,iDur)} = Modal\ Premium_{(iCov,iDur)} - Modal\ Premium\ without\ Extra\ Premium_{(iCov,iDur)}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

8.2.1 Modal Premium

Refer to previous definition

8.2.2 Modal Premium without Extra Premium

$$Modal\ Premium\ without\ Extra\ Premium_{(iCov,iDur)} = ROUND \left\{ Cost\ of\ Insurance_{(iCov,iDur)} * Modal\ factor_{(iCov,iDur)} ; 2 \right\} + ROUND \left\{ Coverage\ fee_{(iCov,iDur)} * Modal\ factor_{(iCov,iDur)} ; 2 \right\}$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection

8.2.2.1 Cost of Insurance Refer to previous definition

8.2.2.2 Modal factor Refer to previous definition

8.2.2.3 Coverage fee Refer to previous definition

9 REPORT: Life Insurance Amount

$$Life\ Insurance\ Amount_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Coverage\ Life\ Insurance\ Amount_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)
→ Coverage 1st Duration

$$Coverage\ Life\ Insurance\ Amount_{(iCov,iDur)} = Coverage\ Life\ Insurance\ Amount\ (from\ Capsil\ at\ first\ but\ can\ be\ modified)$$

→ Where iDur is between Coverage 1st Duration and End of Coverage Projection
→ Additional coverages are not included in the coverages calculated

10 REPORT: Additional Coverages Premium

$$\textit{Additional Coverages Premium}_{(iDur)} = \textit{Fracture Premium}_{(iDur)} + \textit{AD Premium}_{(iDur)} + \textit{AD\&D Premium}$$

→ Where iDur is between Policy 1st Duration and End of Illustration Projection

10.1 Fracture Premium

$$\textit{Fracture Premium}_{(\textit{Policy 1}^{st} \textit{ Duration}+iDur-1)} = \sum_{iCov=1}^{NbCov} \textit{Total Interpolated Premium}_{(iCov, Cov \textit{ 1}^{st} \textit{ Duration}_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.1.1 Total Interpolated Premium

Refer to previous definition

Considering only *Fracture* coverages.

10.2 AD Premium

$$\textit{AD Premium}_{(\textit{Policy 1}^{st} \textit{ Duration}+iDur-1)} = \sum_{iCov=1}^{NbCov} \textit{Total Interpolated Premium}_{(iCov, Cov \textit{ 1}^{st} \textit{ Duration}_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.2.1 Total Interpolated Premium

Refer to previous definition

Considering only *AD* coverages.

10.3 AD&D Premium

AD&D Premium

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.3.1 Total Interpolated Premium

Refer to previous definition

Considering only *AD&D* coverages.

10.4 Accident Premium

$$\textit{Accident Premium}_{(\textit{Policy 1}^{st} \textit{ Duration}+iDur-1)} = \sum_{iCov=1}^{NbCov} \textit{Total Interpolated Premium}_{(iCov, Cov \textit{ 1}^{st} \textit{ Duration}_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.4.1 Total Interpolated Premium

Refer to previous definition

Considering only *Accident* coverages.

10.5 Disability Waiver Premium

$$\textit{Disability Waiver Premium}_{(\textit{Policy 1}^{st} \textit{ Duration}+iDur-1)} = \sum_{iCov=1}^{NbCov} \textit{Total Interpolated Premium}_{(iCov, Cov \textit{ 1}^{st} \textit{ Duration}_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.5.1 Total Interpolated Premium

Refer to previous definition

Considering only *Disability Waiver* coverages.

10.6 GI Premium

$$GI\ Premium_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.6.1 Total Interpolated Premium

Refer to previous definition

Considering only *GI* coverages.

10.7 Bus. Insurability Premium

$$Bus.\ Insurability\ Premium_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.7.1 Total Interpolated Premium

Refer to previous definition

Considering only *Bus. Insurability* coverages.

10.8 Children' s Life Premium

$$Children's\ Life\ Premium_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov,Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.8.1 Total Interpolated Premium

Refer to previous definition

Considering only *Children' s Life* coverages.

10.9 Children' s Fracture Premium

$$Children's\ Fracture\ Premium_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov,Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration

10.9.1 Total Interpolated Premium

Refer to previous definition

Considering only *Children' s Fracture* coverages.

11 REPORT: Coverage Net Premium Reserve

Note: In SIP (Values tab), the variable is named *Coverage Net Premium Reserve*. For presentation purposes in this document, it is referred to as *Total Coverage Net Premium Reserve*.

$$Total\ Coverage\ Net\ Premium\ Reserve_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Coverage\ Net\ Premium\ Reserve_{(iCov,Cov\ 1^{st}\ Duration_{(iCov)}+Coverage\ Dec\ 31^{st}\ Indicator_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration and Coverage Dec 31st Indicator

11.1 Guidelines per coverage

- The *CoverageNetPremiumReserve* must always be calculated for all coverages since their inception.
- For each coverage, the calculation is based on the illustration coverage parameters, which are applied retrospectively to calculate the *CoverageNetPremiumReserve* from inception across all durations.
- The *CoverageNetPremiumReserve* is calculated on the anniversary date of each individual coverage, and not on the policy anniversary date
- For each coverage, it must be determined whether the first duration displayed in the report projection should be deferred. Deferral applies when the policy was issued before December 31, while the coverage was issued after December 31.

$$Coverage\ Net\ Premium\ Reserve_{(iCov, iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Net\ Premium\ Reserve_{(iCov, iInsured, iDur)}$$

→ Where iDur is between 1 and End of Coverage Projection

→ Additional coverages are not included in the coverages calculated

If: $1 \leq iDur \leq MIN\{8; End\ of\ Premium\ Payment_{(iCov)}\} + 1$, **then**

$$Net\ Premium\ Reserve_{(iCov, iInsured, iDur)} = MAX\left\{0; ROUND\left\{\frac{Fiscal\ Insurance\ Amount_{(iCov, iInsured, iDur)}}{100\ 000} * Interpolation\ Duration_{(iCov, iDur)} * Interpolation\ Net\ Premium\ Reserve_{(iCov, iInsured)}; 2\right\}\right\}$$

→ Where iDur is between 1 and End of Coverage Projection

If: $iDur > MIN\{8; End\ of\ Premium\ Payment_{(iCov)}\} + 1$, **then**

$$Net\ Premium\ Reserve_{(iCov, iInsured, iDur)} = MAX\left\{0; ROUND\left\{\frac{Fiscal\ Insurance\ Amount_{(iCov, iInsured, iDur)}}{100\ 000} * \left(TXR_{(iCov, iInsured, iDur)} - \frac{TXR_{(iCov, iInsured, 1)}}{TXP_{(iCov, iInsured, 1)}} * TXP_{(iCov, iInsured, iDur)}\right); 2\right\}\right\}$$

→ Where iDur is between 1 and End of Coverage Projection

11.2 Fiscal Insurance Amount

$$Fiscal\ Insurance\ Amount_{(iCov, iInsured, iDur)} = Input\ Coverage\ Insurance\ Amount$$

→ Where iDur is between 1 and End of Fiscal Projection + 1

11.3 Interpolation Duration

$$Interpolation\ Duration_{(iCov, iDur)} = \frac{iDur - 1}{MIN\{8; End\ of\ Premium\ Payment_{(iCov)}\}}$$

→ Where iDur is between 1 and $MIN(8; End\ of\ Premium\ Payment) + 1$

11.4 Interpolation Net Premium Reserve

$$Interpolation\ Net\ Premium\ Reserve_{(iCov, iInsured)} = MAX\left\{0; TXR_{(iCov, iInsured, MIN\{8; End\ of\ payment\ of\ premiums_{(iCov)}\} + 1)} - \frac{TXR_{(iCov, iInsured, 1)}}{TXP_{(iCov, iInsured, 1)}} * TXP_{(iCov, iInsured, MIN\{8; End\ of\ Premium\ Payment_{(iCov)}\} + 1)}\right\}$$

11.5 TXR

$$TXR_{(iCov, iInsured, iDur)} = \begin{cases} 100\ 000 & \text{if } tPx_{(iCov, iInsured, iDur)} = 0 \\ TRUNC\left\{\frac{Sum\ PVCF\ Benefit_{(iCov, iInsured, iDur)}}{tPx_{(iCov, iInsured, iDur)} * v}; 3\right\} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and End of Coverage Projection

11.5.1 tPx

$$tPx_{(iCov, iInsured, iDur)} = \begin{cases} 1 & \text{if } iDur = 1 \\ TRUNC(tPx_{(iCov, iInsured, iDur-1)} * Px_{(iCov, iInsured, iDur)}; 12) & \text{if } iDur > 1 \end{cases}$$

$$Px_{(iCov, iInsured, iDur)} = 1 - Qx_{(iCov, iInsured, iDur)}$$

→ Where iDur is between 1 and End of Fiscal Projection

11.5.1.1 Qx

$$Qx_{(iCov, iInsured, iDur)} = \begin{cases} 0 & \text{if } iDur = 1 \\ TRUNC\left\{MIN\left\{1; \frac{Mortality\ Table\ Qx_{(iCov, iInsured, iDur)}}{1\ 000} * (1 + Extra\ Premium\ \%_{(iCov, iInsured)}) + \frac{Temporary\ Extra\ Premium\ Rate_{(iCov, iInsured, iDur-1)}}{1\ 000}\right\}; 12\right\} & \text{if } iDur > 1 \end{cases}$$

→ Where iDur is between 1 and End of Fiscal Projection

11.5.1.1.1 Mortality Table Qx

$Mortality\ Table\ Qx_{(iCov,iInsured,iDur)}$ = Prescribed mortality rate from CIA 8692 Mortality Table (*see Product characteristics document*)

→ Where iDur is between 1 and End of CIA8692 Mortality Table (*see Product characteristics document*)

11.5.1.1.2 Extra Premium % Refer to previous definition

11.5.1.1.3 Dollar Extra Premium Rate Refer to previous definition

11.5.2 Sum PVCF Benefit

$$Sum\ PVCF\ Benefit_{(iCov,iInsured,iDur)} = \begin{cases} Cumul\ PVCF\ Benefit_{(iCov,iInsured,End\ of\ Fiscal\ Projection+1)} & \text{if } iDur = 1 \\ Sum\ PVCF\ Benefit_{(iCov,iInsured,1)} - Cumul\ PVCF\ Benefit_{(iCov,iInsured,iDur)} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and End of Coverage Projection

$$Cumul\ PVCF\ Benefit_{(iCov,iInsured,iDur)} = \sum_{i=1}^{iDur} PVCF\ Benefit_{(iCov,iInsured,i)}$$

→ Where iDur is between 1 and End of Fiscal Projection

11.5.2.1 PVCFBenefit

$$PVCF\ Benefit_{(iCov,iInsured,iDur)} = \begin{cases} 0 & \text{if } iDur = 1 \\ TRUNC \left\{ 100\,000 * v_{(iCov,iInsured,iDur)} * tPx_{(iCov,iInsured,iDur-1)} * Qx_{(iCov,iInsured,iDur)} * \left(1 + \frac{Interest\ rate_{(iCov,iInsured,iDur)}}{2} \right); 12 \right\} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and End of Fiscal Projection

11.5.2.1.1 Interest rate

$Interest\ rate_{(iCov,iInsured,iDur)} = 3.5\%$ (**prescribed interest rate**) (*see Product characteristics document*)

→ Where iDur is between 1 and End of Fiscal Projection

11.5.2.1.2 v

$$v_{(iCov,iInsured,iDur)} = ROUND \left\{ \left(\frac{1}{1 + Interest\ rate_{(iCov,iInsured,iDur)}} \right)^{iDur-1}; 12 \right\}$$

→ Where iDur is between 1 and End of Fiscal Projection + 1

11.5.2.1.3 tPx Refer to previous definition

11.5.2.1.4 Qx Refer to previous definition

11.6 TXP

$$TXP_{(iCov,iInsured,iDur)} = \begin{cases} 0 & \text{if } tPx_{(iCov,iInsured,iDur)} = 0 \\ TRUNC \left\{ \frac{Sum\ PVCF\ Premium_{(iCov,iInsured,iDur)}}{tPx_{(iCov,iInsured,iDur)} * v_{(iCov,iInsured,iDur)}}; 3 \right\} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and End of Coverage Projection

11.6.1 tPx

Refer to previous definition

11.6.2 v

Refer to previous definition

11.6.3 Sum PVCF Premium

$$Sum\ PVCF\ Premium_{(iCov,iInsured,iDur)} = \begin{cases} Cumul\ PVCF\ Premium_{(iCov,iInsured,End\ of\ Projection)} & \text{if } iDur = 1 \\ Sum\ PVCF\ Premium_{(iCov,iInsured,1)} - Cumul\ PVCF\ Premium_{(iCov,iInsured,iDur-1)} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and End of Coverage Projection

$$Cumul\ PVCF\ Premium_{(iCov,iInsured,iDur)} = \sum_{i=1}^{iDur} PVCF\ Premium_{(iCov,iInsured,i)}$$

11.6.3.1 PVCF Premium

$$\begin{aligned}
 PVCF \text{ Premium}_{(iCov,iInsured,iDur)} = & TRUNC \left\{ \left[\left(\left(Premium \text{ rate}_{(iCov,iInsured,iDur)} + Extra \text{ premium rate}_{(iCov,iInsured,iDur)} * Extra \text{ premium } \%_{(iCov,iInsured)} \right) * PREM_ADJ\%_{(iCov,iDur)} \right. \right. \\
 & + Temporary \text{ Extra Premium Rate}_{(iCov,iInsured,iDur)} \left. \right) * 100 \\
 & + TRUNC \left\{ \frac{PREM_ADJ\$_{(iCov,iDur)} * 100\,000}{Fiscal \text{ Insurance Amount}_{(iCov,iInsured,iDur)}}; 2 \right\} \\
 & + TRUNC \left\{ \frac{Coverage \text{ fee}_{(iCov,iDur)} * 100\,000}{Fiscal \text{ Insurance Amount}_{(iCov,iInsured,iDur)}}; 2 \right\} \left. \right] \\
 & * v_{(iCov,iInsured,iDur)} * tPx_{(iCov,iInsured,iDur)}; 12 \left. \right\}
 \end{aligned}$$

→ Where iDur is between 1 and End of Coverage Projection

11.6.3.1.1 Premium Rate Refer to previous definition

11.6.3.1.2 Extra Premium Rate Refer to previous definition

11.6.3.1.3 Extra Premium % Refer to previous definition

11.6.3.1.4 PREM_ADJ% Refer to previous definition

11.6.3.1.5 PREM_ADJ\$ Refer to previous definition

11.6.3.1.6 Dollar Extra Premium Rate Refer to previous definition

11.6.3.1.7 Coverage fee Refer to previous definition

11.6.3.1.8 Fiscal Insurance Amount Refer to previous definition

12 REPORT: Net Amount at Risk (NAAR)

$$Net \text{ Amount at Risk}_{(Policy \ 1^{st} \ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Coverage \ Net \ Amount \ at \ Risk_{(iCov, \ Cov \ 1^{st} \ Duration_{(iCov)}+Coverage \ Dec \ 31^{st} \ Indicator_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration and Coverage Dec 31st Indicator

12.1 Coverage Net Amount at Risk

$$Coverage \ Net \ Amount \ at \ Risk_{(iCov,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Coverage \ Net \ Amount \ at \ Risk_{(iCov,iInsured, \ iDur)}$$

$$Coverage \ Net \ Amount \ at \ Risk_{(iCov,iInsured,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} (Fiscal \ Insurance \ Amount_{(iCov,iInsured,iDur)} - Net \ Premium \ Reserve_{(iCov,iInsured,iDur)})$$

→ Where iDur is between 1 and End of Coverage Projection

→ Additional coverages are not included in the coverages calculated

12.1.1 Fiscal Insurance Amount

Refer to previous definition

12.1.2 Net Premium Reserve

Refer to previous duration

13 REPORT: Net Cost of pure Insurance (NCPI)

⇒ If $iDur = 1$, then

$$Net\ Cost\ of\ pure\ Insurance_{(Policy\ 1^{st}\ Duration+iDur-1)} = \begin{cases} 0 & \text{if } Policy\ Dec\ 31^{st}\ Indicator = 1 \\ \sum_{iCov=1}^{NbCov} Coverage\ Net\ Cost\ of\ pure\ Insurance_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+Coverage\ Dec\ 31^{st}\ Indicator_{(iCov)}+iDur-1)} & \text{otherwise} \end{cases}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration and Coverage Dec 31st Indicator

⇒ If $iDur > 1$, then

$$Net\ Cost\ of\ pure\ Insurance_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Coverage\ Net\ Cost\ of\ pure\ Insurance_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+Coverage\ Dec\ 31^{st}\ Indicator_{(iCov)}+iDur-1)}$$

→ Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)

→ Coverage 1st Duration and Coverage Dec 31st Indicator

13.1 Coverage Net Cost of pure Insurance

$$Coverage\ Net\ Cost\ of\ pure\ Insurance_{(iCov,iDur)} = \sum_{iInsured=1}^{NbInsuredOnCov} Coverage\ Net\ Cost\ of\ pure\ Insurance_{(iCov,iInsured,iDur)}$$

$$Coverage\ Net\ Cost\ of\ pure\ Insurance_{(iCov,iInsured,iDur)} = ROUND \left\{ Qx_{(iCov,iInsured,iDur+1)} * Coverage\ Net\ Amount\ at\ Risk_{(iCov,iInsured,iDur)} ; 2 \right\}$$

→ Where iDur is between 1 and End of Coverage Projection

→ Additional coverages are not included in the coverages calculated

13.1.1 Qx

Refer to previous definition

13.1.2 Coverage Net Amount at Risk

Refer to previous definition

14 REPORT: Cumulative NCPI

$$Cumulative\ NCPI_{(iDur)} = \begin{cases} Policy\ Cumulative\ NCPI & \text{if } iDur = Policy\ 1^{st}\ Duration\ AND\ Policy\ Dec\ 31^{st}\ Indicator = 1 \\ Policy\ Cumulative\ NCPI + Net\ Cost\ of\ pure\ Insurance_{(iDur)} & \text{if } iDur = Policy\ 1^{st}\ Duration\ AND\ Policy\ Dec\ 31^{st}\ Indicator = 0 \\ Cumulative\ NCPI_{(iDur-1)} + Net\ Cost\ of\ pure\ Insurance_{(iDur)} & \text{if } iDur > Policy\ 1^{st}\ Duration \end{cases}$$

→ Where iDur is between Policy 1st Duration and End of Illustration Projection

14.1 Policy Cumulative NCPI

Refer to previous definition

14.2 Policy 1st Duration

Refer to previous definition

14.3 Policy Dec 31st Indicator

Refer to previous definition

14.4 Net Cost of pure Insurance

Refer to previous definition

15 **REPORT: ACB Coverage Cost**

$$ACB\ Coverage\ Cost_{(Policy\ 1^{st}\ Duration+iDur-1)} = \sum_{iCov=1}^{NbCov} Total\ Interpolated\ Premium_{(iCov,\ Cov\ 1^{st}\ Duration_{(iCov)}+iDur-1)}$$

- Where iDur is between 1 and (End of Illustration Projection - Policy 1st Duration)
- Coverage 1st Duration
- Only life insurance coverages are included in this calculation

15.1 **Total Interpolated Premium**

Refer to previous definition
Considering only life insurance coverages.

16 **REPORT: Cumulative ACB Coverage Cost**

$$Cumulative\ ACB\ Coverage\ Cost_{(iDur)} = \begin{cases} Policy\ ACB + Policy\ Cumulative\ NCPI + ACB\ Coverage\ Cost_{(iDur)} & \text{if } iDur = Policy\ 1^{st}\ Duration \\ Cumulative\ ACB\ Coverage\ Cost_{(iDur-1)} + ACB\ Coverage\ Cost_{(iDur)} & \text{if } iDur > Policy\ 1^{st}\ Duration \end{cases}$$

- Where iDur is between Policy 1st Duration and End of Illustration Projection
- Additional coverages are not included in the coverages calculated

16.1 **ACB Coverage Cost**

Refer to previous definition

16.2 **Policy ACB**

Refer to previous definition

16.3 **Policy Cumulative NCPI**

Refer to previous definition

16.4 **Policy 1st Duration**

Refer to previous definition

17 **REPORT: Adjusted Cost Basis (ACB)**

$$ACB_{(iDur)} = MAX \left\{ 0 ; ACB\ Before\ MAX_{(iDur)} \right\}$$

$$ACB\ Before\ MAX_{(iDur)} = \begin{cases} Policy\ ACB + ACB\ Coverage\ Cost_{(iDur)} - Net\ Cost\ of\ pure\ Insurance_{(iDur)} & \text{if } iDur = Policy\ 1^{st}\ Duration \\ ACB\ Before\ MAX_{(iDur-1)} + ACB\ Coverage\ Cost_{(iDur)} - Net\ Cost\ of\ pure\ Insurance_{(iDur)} & \text{if } iDur > Policy\ 1^{st}\ Duration \end{cases}$$

- Where iDur is between Policy 1st Duration and End of Illustration Projection

17.1 **ACB Coverage Cost**

Refer to previous definition

17.2 **Net Cost of pure Insurance**

Refer to previous definition

17.3 **Policy ACB**

Refer to previous definition

17.4 **Policy 1st Duration**

Refer to previous definition